

Finalized columns and correct names:

1. YEAR
2. MONTH
3. DAY_OF_MONTH
4. DAY_OF_WEEK
5. FL_DATE
6. OP_UNIQUE_CARRIER
7. OP_CARRIER_AIRLINE_ID
8. TAIL_NUM
9. OP_CARRIER_FL_NUM
10. ORIGIN_AIRPORT_ID
11. ORIGIN
12. ORIGIN_CITY_NAME
13. ORIGIN_STATE_ABR
14. ORIGIN_STATE_NM
15. ORIGIN_WAC
16. DEST_AIRPORT_ID
17. DEST
18. DEST_CITY_NAME
19. DEST_STATE_ABR
20. DEST_STATE_NM
21. DEST_WAC
22. CRS_DEP_TIME
23. DEP_TIME
24. DEP_DELAY
25. DEP_DELAY_NEW
26. DEP_DEL15
27. TAXI_OUT
28. WHEELS_OFF
29. WHEELS_ON
30. TAXI_IN
31. CRS_ARR_TIME
32. ARR_TIME
33. ARR_DELAY
34. ARR_DELAY_NEW
35. ARR_DEL15
36. CANCELLED
37. CANCELLATION_CODE

38. DIVERTED

39. CRS_ELAPSED_TIME

40. ACTUAL_ELAPSED_TIME

41. AIR_TIME

42. DISTANCE

43. CARRIER_DELAY

44. WEATHER_DELAY

45. NAS_DELAY

46. SECURITY_DELAY

47. LATE_AIRCRAFT_DELAY

48. DIV_AIRPORT_LANDINGS

49. DIV_REACHED_DEST

50. DIV_ACTUAL_ELAPSED_TIME

51. DIV_ARR_DELAY

52. DIV_DISTANCE

53. DIV_AIRPORT

54. DIV_WHEELS_ON

55. DIV_WHEELS_OFF

56. DIV_TOTAL_GTIME

Step 0: Understand the Meaning

One row in the dataset represents

- One airline flight occurrence
- On one date
- Between one origin airport
- And one destination airport

The row currently mixes several entities and informations:

- Airline
- Airport
- State
- Date
- Flight
- Performance
- Diversion
- Cancellation

Step 1: Function Dependencies

- Airline
 - OP_CARRIER_AIRLINE_ID → OP_UNIQUE_CARRIER
 - From lookup file: OP_CARRIER_AIRLINE_ID → AIRLINE_NAME (from lookup)
- Airport
- Origin Airport
 - ORIGIN_AIRPORT_ID → ORIGIN, ORIGIN_CITY_NAME, ORIGIN_STATE_ABR, ORIGIN_STATE_NM, ORIGIN_WAC
- Destination Airport
 - DEST_AIRPORT_ID → DEST, DEST_CITY_NAME, DEST_STATE_ABR, DEST_STATE_NM, DEST_WAC
- Date
 - FL_DATE → YEAR, MONTH, DAY_OF_MONTH, DAY_OF_WEEK
- State
 - STATE_ABR → STATE_NM (from lookup)
- Cancellation
 - CANCELLATION_CODE → CANCELLATION_DESCRIPTION (from lookup)
- Route
 - (ORIGIN_AIRPORT_ID, DEST_AIRPORT_ID) → DISTANCE

Step 2: 1NF

The BTS flight dataset already satisfies 1NF because each row represents one flight occurrence and all attributes contain atomic values.

Step 3: 2NF

To eliminate redundancy, airline, airport, route and date information are separated into their own relations.

```
AIRLINE(  
    OP_CARRIER_AIRLINE_ID PK,  
    OP_UNIQUE_CARRIER,  
    AIRLINE_NAME  
)
```

Since origin and destination airports describe the same thing as airports in general, we can normalize them into one where they reuse the same entity, and they reference the following table:

```
AIRPORT(  
    AIRPORT_ID PK,  
    AIRPORT_CODE,  
    AIRPORT_NAME,  
    CITY_NAME,  
    STATE_ABR,
```

```

        STATE_NAME,
        WAC
    )

DATE_DIMENSION(
    FL_DATE PK,
    YEAR,
    MONTH,
    DAY_OF_MONTH,
    DAY_OF_WEEK
)

ROUTE(
    ROUTE_ID PK,
    ORIGIN_AIRPORT_ID FK,
    DEST_AIRPORT_ID FK,
    DISTANCE
)

```

ROUTE_ID is introduced as a surrogate key to uniquely identify each origin-destination route and serves as the primary key of the ROUTE table.

```

FLIGHT(
    FLIGHT_ID PK,
    OP_CARRIER_AIRLINE_ID FK,
    FL_DATE FK,
    ROUTE_ID FK,
    OP_CARRIER_FL_NUM,
    TAIL_NUM
)

```

FLIGHT_ID is introduced as a surrogate key to uniquely identify each flight occurrence and serves as the primary key of the FLIGHT table.

Step 4: 3NF

Now we remove transitive dependencies.

AIRPOR_ID determines STATE_ABR, and STATE_ABR determines STATE_NAME. Therefore, STATE_NAME is transitively dependent on AIRPORT_ID. So:

```

STATE (
    STATE_ABR PK,
    STATE_NAME
)

```

And AIRPORT becomes:

```

AIRPORT(
    AIRPORT_ID PK,

```

```
AIRPORT_CODE,  
AIRPORT_NAME,  
CITY_NAME,  
STATE_ABR FK,  
WAC  
)
```

Since CANCELLATION_CODE → CANCELLATION_DESCRIPTION:

```
CANCELLATION_REASON(  
    CANCELLATION_CODE PK,  
    DESCRIPTION  
)
```

The diversion attributes only apply when DIVERTED = 1. Most flights have DIVERTED = 0, so:

```
FLIGHT_DIVERSION(  
    FLIGHT_ID PK/FK,  
    DIV_AIRPORT_LANDINGS,  
    DIV_REACHED_DEST,  
    DIV_ACTUAL_ELAPSED_TIME,  
    DIV_ARR_DELAY,  
    DIV_DISTANCE,  
    DIV_AIRPORT,  
    DIV_WHEELS_ON,  
    DIV_WHEELS_OFF,  
    DIV_TOTAL_GTIME  
)
```

Everything operational stays in the flight performance table:

```
FLIGHT_PERFORMANCE(  
    FLIGHT_ID PK/FK,  
    CRS_DEP_TIME,  
    DEP_TIME,  
    DEP_DELAY,  
    DEP_DELAY_NEW,  
    DEP_DEL15,  
    TAXI_OUT,  
    WHEELS_OFF,  
    WHEELS_ON,  
    TAXI_IN,  
    CRS_ARR_TIME,  
    ARR_TIME,  
    ARR_DELAY,  
    ARR_DELAY_NEW,  
    ARR_DEL15,
```

```

    CANCELLED,
    DIVERTED,
    CANCELLATION_CODE FK,
    CRS_ELAPSED_TIME,
    ACTUAL_ELAPSED_TIME,
    AIR_TIME,
    CARRIER_DELAY,
    WEATHER_DELAY,
    NAS_DELAY,
    SECURITY_DELAY,
    LATE_AIRCRAFT_DELAY
)

```

Final Schema:

```

AIRLINE(
    OP_CARRIER_AIRLINE_ID PK,
    OP_UNIQUE_CARRIER,
    AIRLINE_NAME
)

```

Stores carrier information once instead of repeating it for every flight.

```

STATE(
    STATE_ABR PK,
    STATE_NAME
)

```

Removes transitive dependency. Source columns are origin and destination STATE_ABR and STATE_NM.

```

AIRPORT(
    AIRPORT_ID PK,
    AIRPORT_CODE,
    AIRPORT_NAME,
    CITY_NAME,
    STATE_ABR FK,
    WAC
)

```

Stores airport information once from the origin and destination source columns.

```

DATE_DIMENSION(
    FL_DATE PK,
    YEAR,
    MONTH,
    DAY_OF_MONTH,
)

```

```
        DAY_OF_WEEK
    )
    Stores date attributes once.
```

```
CANCELLATION_REASON(
    CANCELLATION_CODE PK,
    DESCRIPTION
)
    Stores cancellation code descriptions.
```

```
ROUTE(
    ROUTE_ID PK,
    ORIGIN_AIRPORT_ID FK,
    DEST_AIRPORT_ID FK,
    DISTANCE
)
    Stores origin-destination routes. DISTANCE is stored in ROUTE because it depends on the
    origin-destination pair rather than an individual flight occurrence.
```

```
FLIGHT(
    FLIGHT_ID PK,
    OP_CARRIER_AIRLINE_ID FK,
    FL_DATE FK,
    ROUTE_ID FK,
    OP_CARRIER_FL_NUM,
    TAIL_NUM
)
    Stores identifying flight information
```

```
FLIGHT_PERFORMANCE(
    FLIGHT_ID PK/FK,
    CRS_DEP_TIME,
    DEP_TIME,
    DEP_DELAY,
    DEP_DELAY_NEW,
    DEP_DEL15,
    TAXI_OUT,
    WHEELS_OFF,
    WHEELS_ON,
    TAXI_IN,
    CRS_ARR_TIME,
    ARR_TIME,
    ARR_DELAY,
    ARR_DELAY_NEW,
```

```
ARR_DEL15,  
CANCELLED,  
DIVERTED,  
CANCELLATION_CODE FK,  
CRS_ELAPSED_TIME,  
ACTUAL_ELAPSED_TIME,  
AIR_TIME,  
CARRIER_DELAY,  
WEATHER_DELAY,  
NAS_DELAY,  
SECURITY_DELAY,  
LATE_AIRCRAFT_DELAY
```

)

Stores all operational and delay metrics.

```
FLIGHT_DIVERSION(  
    FLIGHT_ID PK/FK,  
    DIV_AIRPORT_LANDINGS,  
    DIV_REACHED_DEST,  
    DIV_ACTUAL_ELAPSED_TIME,  
    DIV_ARR_DELAY,  
    DIV_DISTANCE,  
    DIV_AIRPORT,  
    DIV_WHEELS_ON,  
    DIV_WHEELS_OFF,  
    DIV_TOTAL_GTIME
```

)

Stores information only for diverted flights. This is separate because most flights have DIVERTED = 0 and all diversion columns are NULL. So separating them eliminates unnecessary NULLs.

There are a total of 9 normalized tables with 56 attributes in 3NF